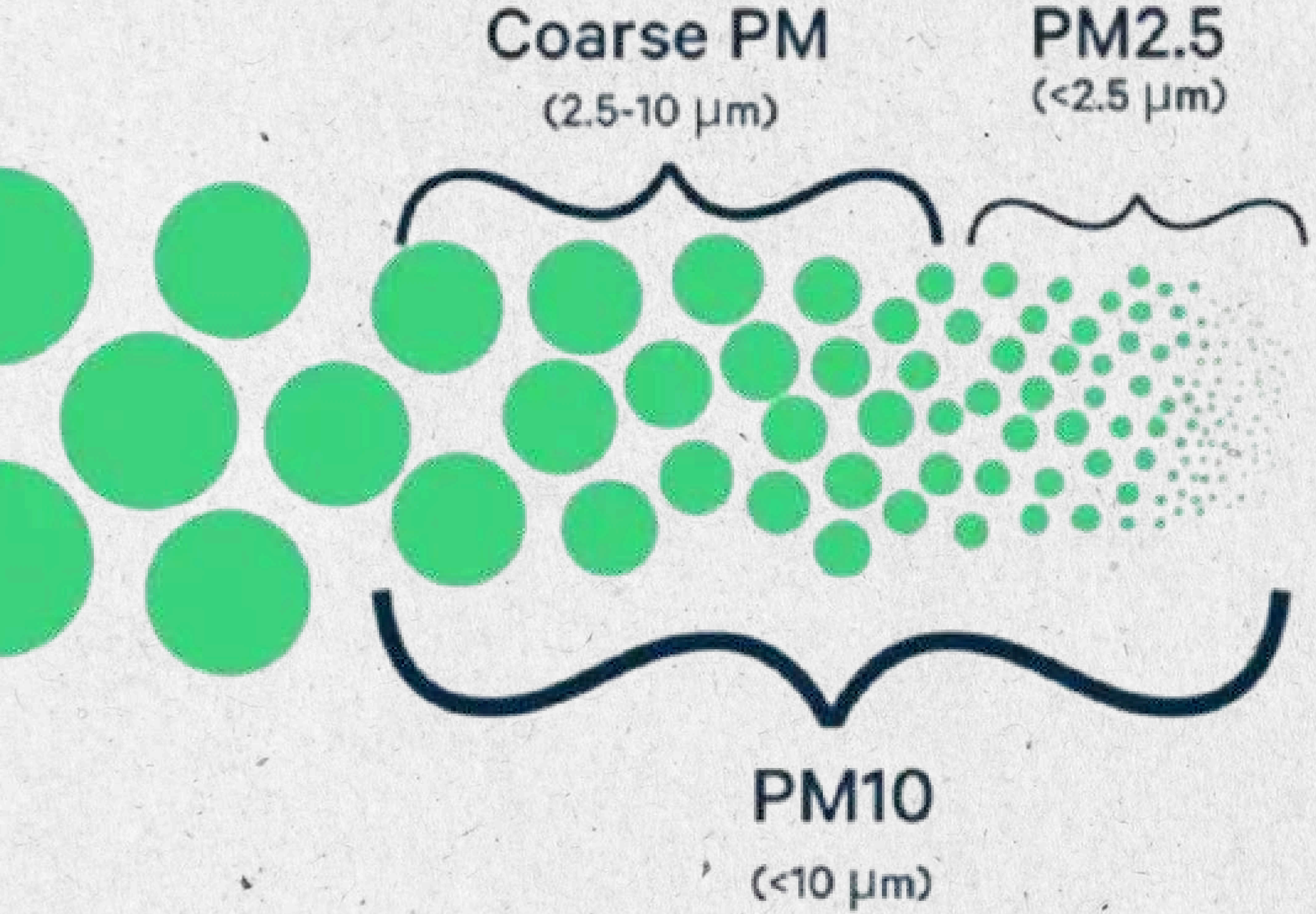
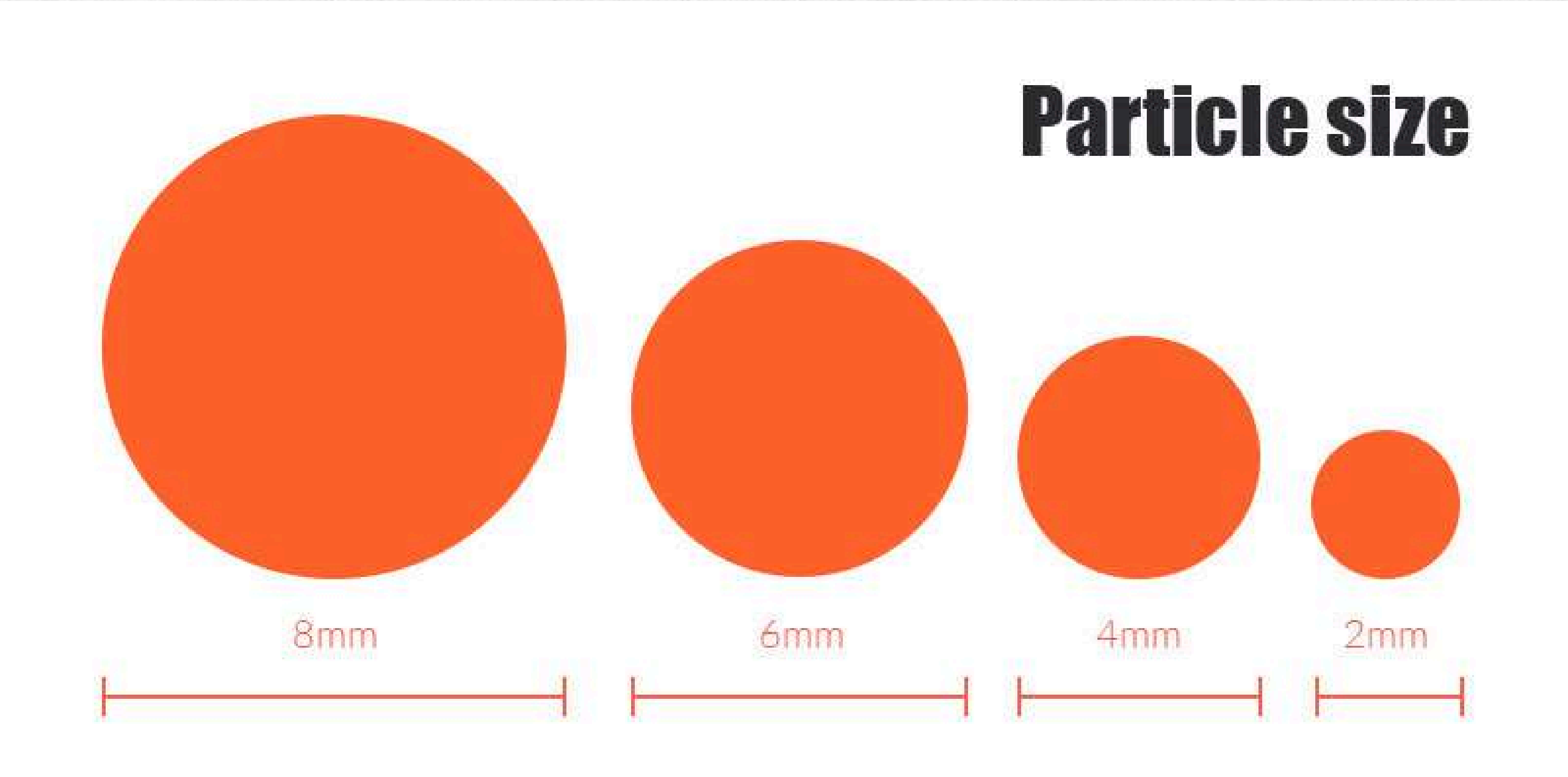


GENERALIZED PARTICLE MORPHOLOGY LEARNING FROM DIGITAL IMAGES

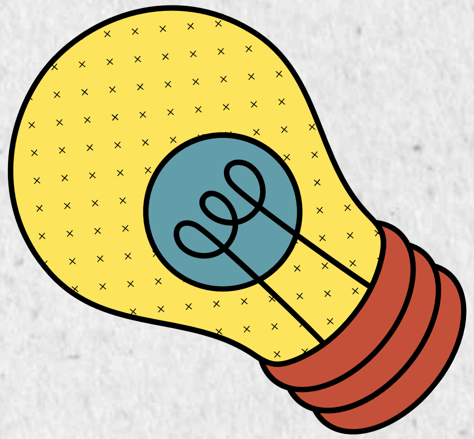
MorphoVision-MVS

PROBLEM STATEMENT

Materials and pharmaceutical researchers need to analyze microparticle morphology from microscopy images (SEM, Brightfield, Fluorescence) to characterize particle size, shape, and quality.



Particle geometry analysis is essential because properties like size, shape, and surface characteristics directly influence how particles behave in materials and pharmaceutical applications. It helps ensure product quality, consistency, and effectiveness by predicting flow, reactivity, and stability.



RESEARCH MOTIVATION

1. Pharmaceutical Research & Development
   particle size affects how drugs work in the body.
2. Materials Science & Engineering
   - helps understand material strength and quality.
3. Quality Control & Manufacturing
   - Ensures product quality by analyzing particles quickly
4. Nanotechnology
   - Studies very small particles for research and industrial use
5. Biotechnology & Life Sciences
   - Analyzes biological particles like cells and microorganisms

RESEARCH & DEVELOPMENT OVERVIEW

Team Members & Work Division

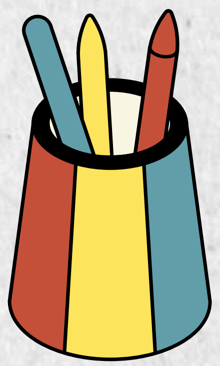
Image Processing Module: Focused on particle detection, segmentation, and noise removal techniques.

UI/UX Design: Developed an intuitive interface for uploading images and visualizing results.

Metrics Implementation: Worked on calculating particle properties like size, shape, circularity, and surface features.

Testing & Validation: Evaluated accuracy using different datasets and ensured reliability of outputs.

LITERATURE SURVEY

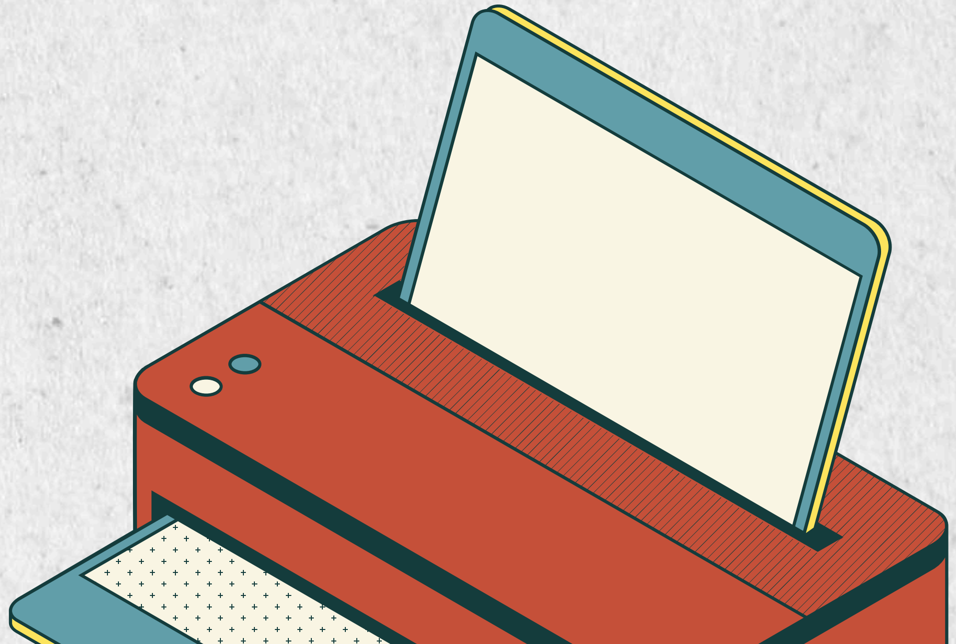


What Already Exists — And What's Missing

- ImageJ (NIH): Open-source, powerful — but requires manual configuration for every image type; computes basic metrics only; no pipeline automation
- MATLAB Image Processing Toolbox: Very accurate — but costly, requires programming expertise, not accessible for non-CS researchers
- Commercial Tools (Malvern, Sympatec): Expensive hardware-dependent; cannot work from existing digital images alone

Key Gaps Found in Literature:

1. No tool computes Heywood Circularity Factor and PA Fractal Dimension together in a pipeline
2. No automatic imaging mode detection (SEM vs Brightfield vs Fluorescence)
3. No one-click pipeline from image → full report
4. No morphological classification based on combined multi-metric rules



PROPOSED SOLUTION: MORPHOVISION-MVS

A specialized Java-based desktop research dashboard that transforms any SEM, Brightfield, or Fluorescence microscopy image into a complete quantitative morphological analysis — automatically.

Material-Independent — works on any particle image regardless of material
Mode-Aware — auto-tunes processing parameters per imaging type
One-Click Pipeline — no manual configuration needed
14+ Metrics — goes far beyond what standard tools provide

Language: Java (SE 11+)
Base Engine: ImageJ / ImageJ2 (NIH Open Source)
UI Framework: Java Swing — custom MorphoDesktop MDI interface
Charting: Java2D — zero external dependencies, publication-quality plots
Output: CSV (per-particle data)

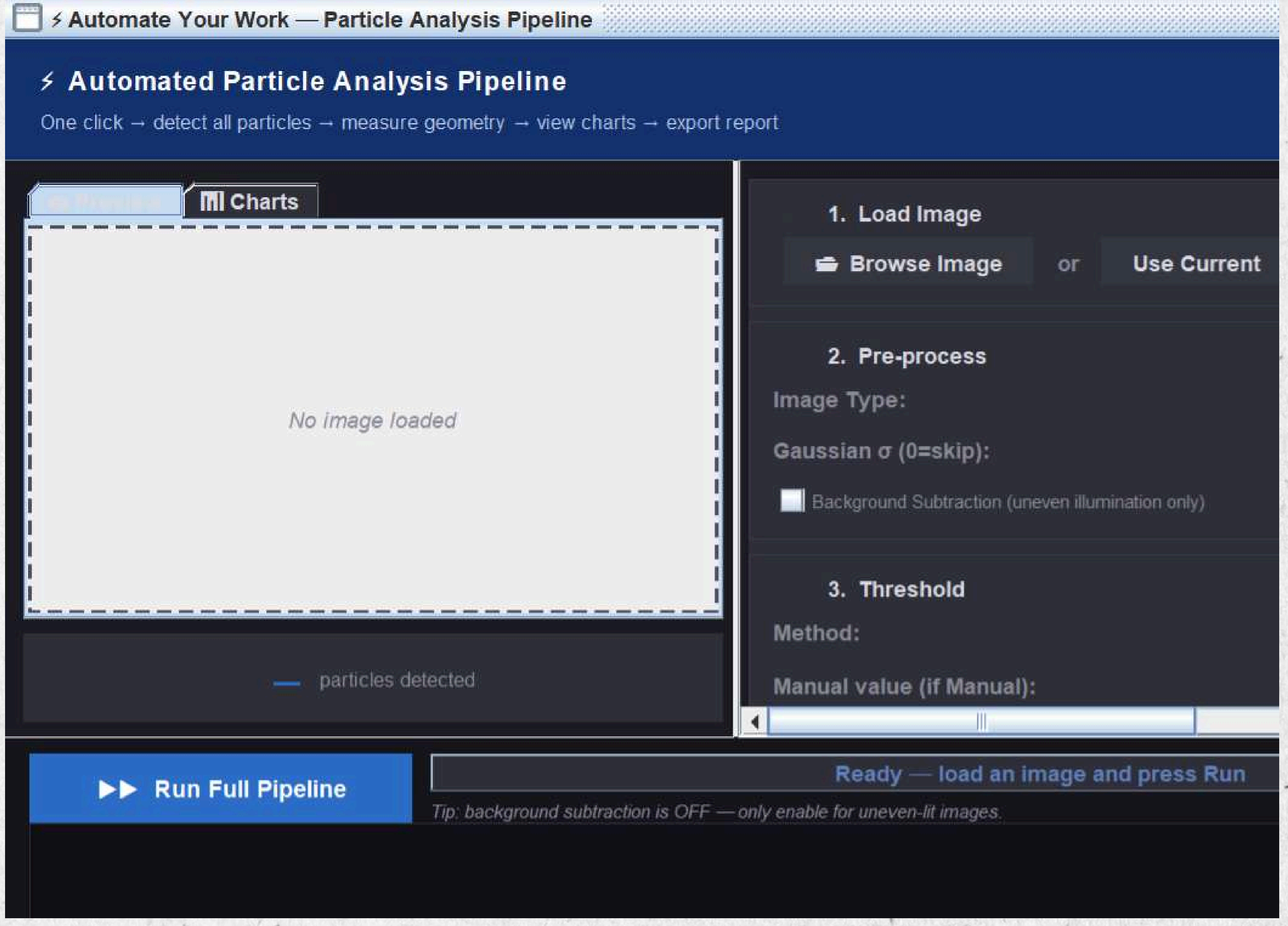
THE MORPHOLOGICAL METRICS

Metric	Formula	Meaning
Area	Pixel count × $scale^2$	Particle size
Perimeter	Boundary length	Outline length
Circularity	$4\pi A / P^2$	1 = perfect circle, 0 = irregular
Equivalent Diameter	$2\sqrt{(A/\pi)}$	Diameter of circle with same area
Aspect Ratio	Major / Minor axis	Elongation ratio
Feret Max / Min	Max & min caliper	Widest and narrowest span
Solidity	Area / Convex Hull Area	Detects agglomerates (concave outline)
Roundness	$4A / (\pi \times Major^2)$	Distinguishes circles from ellipses
Heywood Circularity Factor	$P / (\pi \times EquivDiam)$	Pharma standard — 1 = circle, >1 = irregular
Elongation Index	(Major − Minor) / (Major + Minor)	0 = circle, 1 = perfect line/fiber

ALGORITHMS USED: MEASUREMENT & CLASSIFICATION

#	Algorithm	Stage	Used For
1	Gaussian Blur	Pre-process	Noise removal, smoothing
2	Rolling Ball BG Subtraction	Pre-process	Uneven illumination correction
3	Otsu Thresholding	Segmentation	Brightfield / Fluorescence images
4	Triangle Thresholding	Segmentation	SEM images
5	Watershed	Segmentation	Separation of touching particles
6	Connected Component Labeling	Detection	Particle identification
7	Ellipse Fitting	Measurement	Aspect Ratio, Roundness,
8	Convex Hull	Measurement	Solidity, Agglomerate detection
9	PA Fractal Dimension	Measurement	Surface roughness metric
10	Heywood Factor	Measurement	Pharmaceutical circularity
11	Rule-Based Classifier	AI Logic	Morphological auto-labeling

AUTOMATED PIPELINE ARCHITECTURE



Step 1: Image Input & Preprocessing

Step 2: Automatic Particle Detection

Step 3: Morphological Measurement

Step 4: Real-time Analytics & Visualization

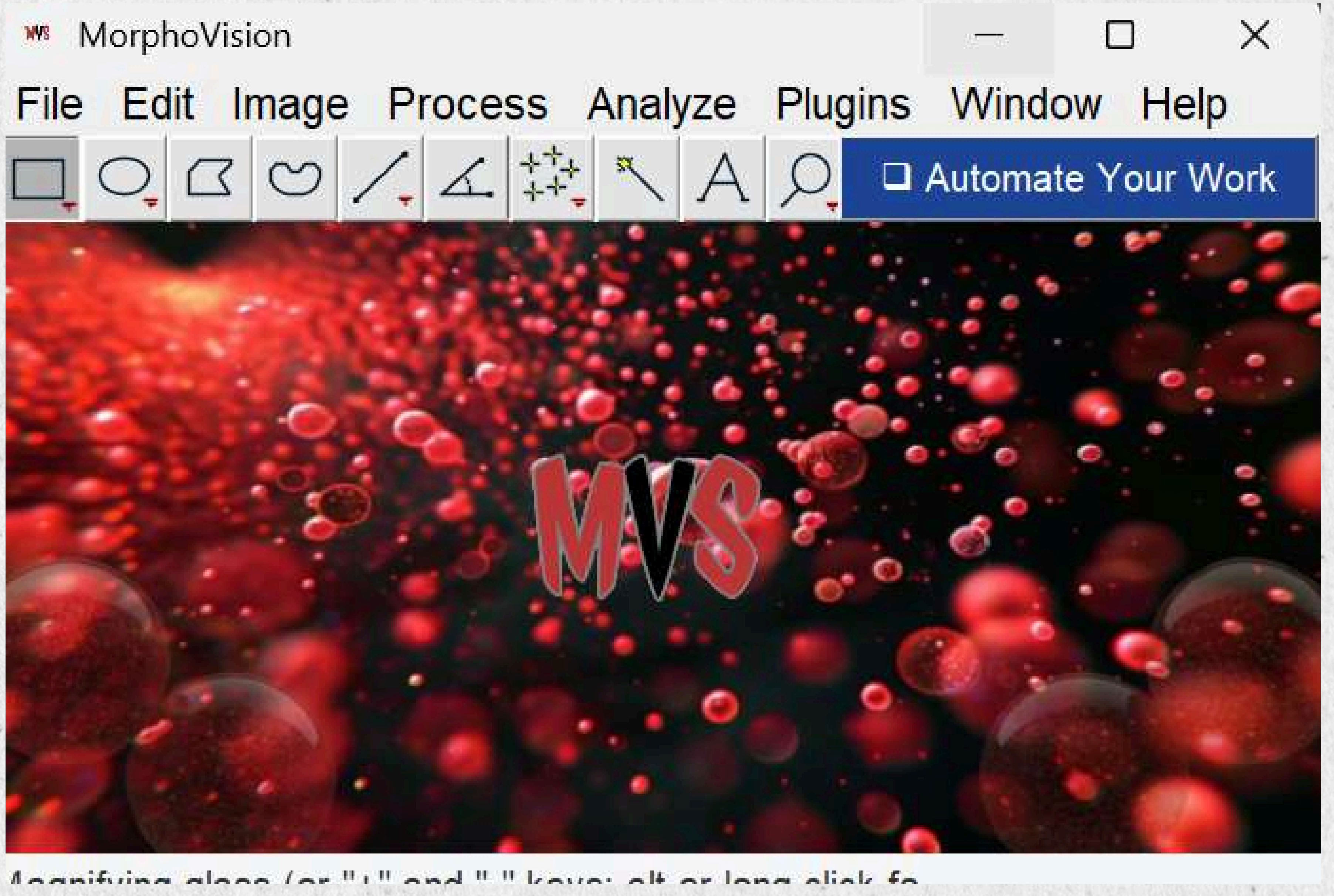
The system generates ready-to-use reports and exports data for further analysis.

ANALYTICS & VISUALIZATION OUTPUTS

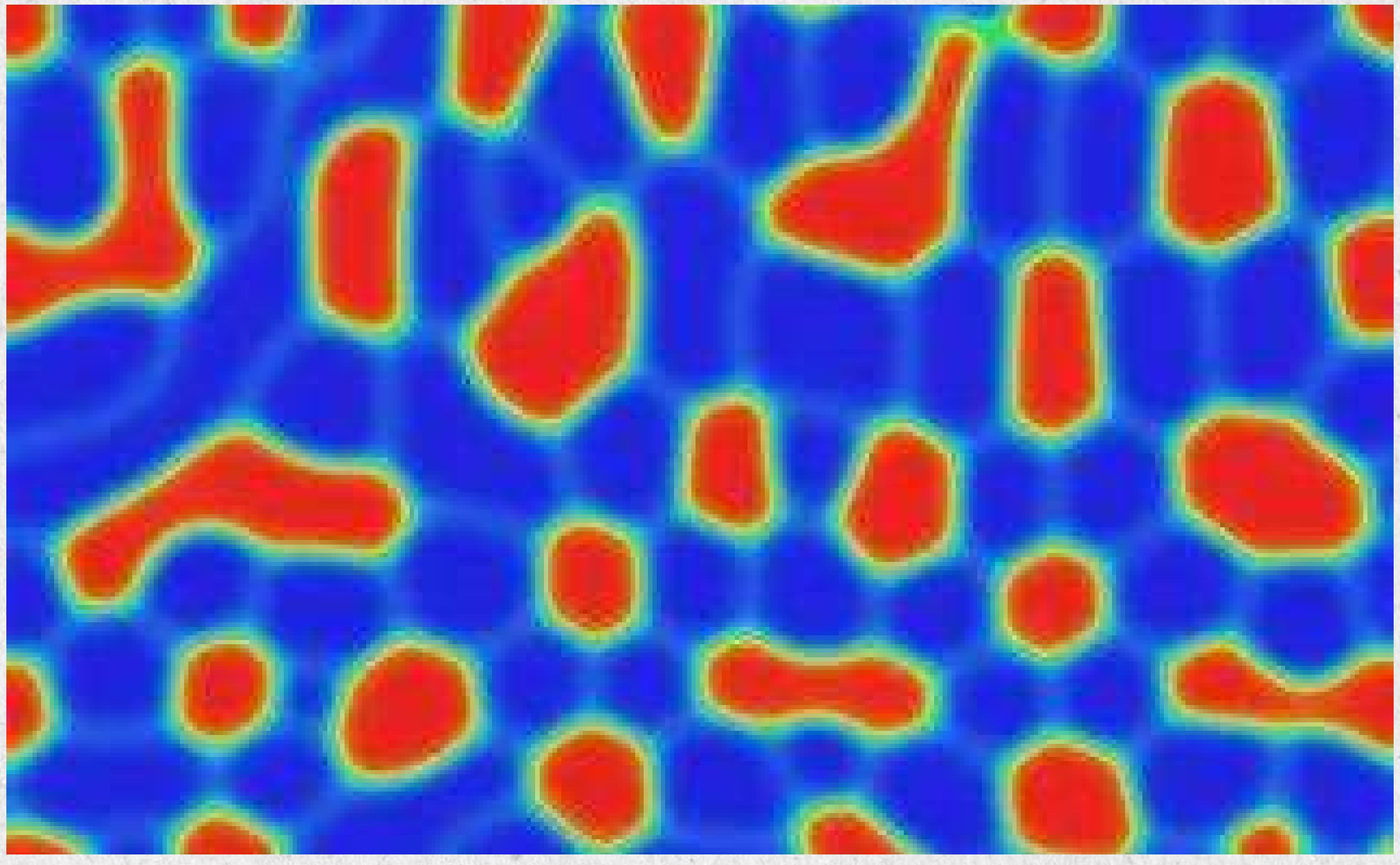
#	Visualization	Description
1	Particle Size Distribution	Histogram showing particle size frequency; helps identify uniformity and outliers
2	Area vs Perimeter Plot	Scatter plot showing relation between area and perimeter; reveals shape and compactness
3	Circularity Distribution	Histogram of Heywood Circularity Factor; indicates how close particles are to circular
4	Elongation Ratio Distribution	Histogram showing elongation patterns; identifies fibers and elongated particles
5	Solidity Distribution	Shows particle solidity; helps detect irregular shapes and agglomerates
6	Fractal Dimension Distribution	Histogram of fractal values; measures surface roughness of particles
7	Classification Breakdown	Pie/bar chart showing counts of different particle types (spherical, irregular, etc.)
8	Statistical Summary Dashboard	Displays mean, median, standard deviation, min/max, and percentile values

UNIQUENESS OF THE PROJECT

- Interactive Analytics Dashboard
- Advanced Metrics
- One-Click Automation & 8 Real-Time Analytics Charts
- Universal Material Independence (Multi-Modal Support)
- No Programming Required



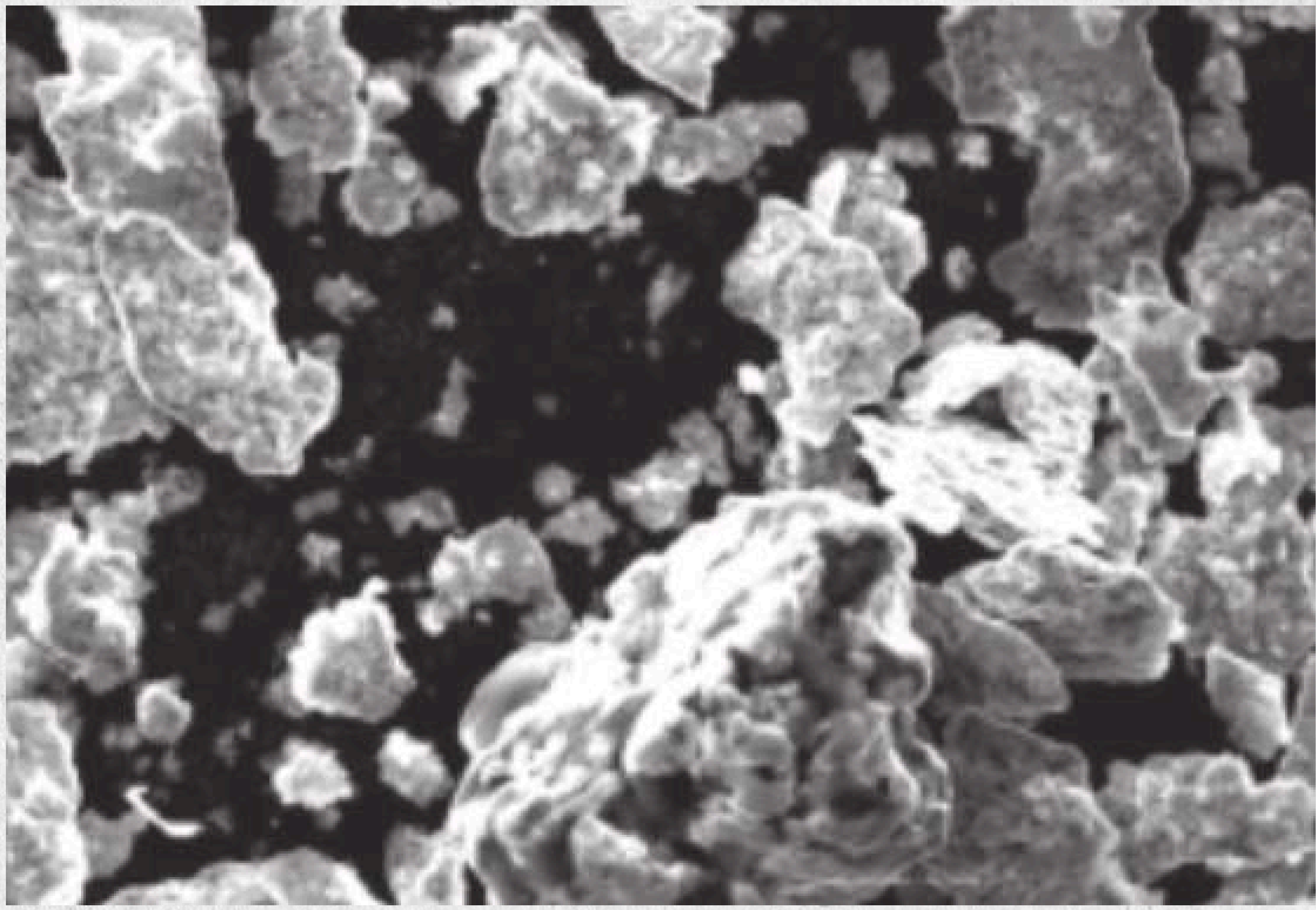
CHALLENGES WE FACED



1. Particle Segmentation Was Highly Unstable in Real Images
2. SEM Images Introduced Specular Highlights
3. Background Detection Varied Across Image Types
4. Touching & Agglomerated Particles Were Difficult to Separate
5. Pixel-Based Measurements Lacked Real-World Meaning

FUTURE SCOPE

1. Machine Learning & AI Enhancement
2. 3D Particle Analysis Support
3. Cloud-Based Analytics
4. Agglomerative Particle Detection & Separation
5. Particle Depth & Z-Stack Analysis



CONCLUSION

MorphoVision-MVS successfully addresses the research gap in generalized particle morphology analysis. It bridges the gap between raw microscopy images and quantitative research data, reducing analysis time from hours to seconds while providing richer morphological insights than any comparable open-source tool.

THANK YOU